Pushkar Raj Wadhawan

ch24btech11030@iith.ac.in | pushkarrajwadhawan@gmail.com | +91 6284118034 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.60
SOHRAB PUBLIC SCHOOL NABHA ROAD MALERKOTLA PB	Class XII (CBSE)	2024	90.40%
SVM DHURI ROAD TEH MALERKOTLA DT SANGRUR PB	Class X (CBSE)	2022	95.60%

SKILLS

- **Languages**: C, C++, Python, JavaScript, SQL
- **Core CS**: Data Structures & Algorithms, Object-Oriented Programming
- **Developer Tools:** Git, GitHub, MySQL
- **Relevant Courses:** Introduction to Programming in C, Intro to AI/ML, Numerical Methods, Linear Algebra

PROJECTS

Chess Engine | *C++, Object-Oriented Design*

- Implemented a chess engine with an **abstract Piece base class** and **6 derived piece types** - pawn, knight, bishop, rook, queen, king, each generating its own legal moves through a shared move-generation framework.
- Handled **castling**, **en passant,** and **pawn promotion** as special cases layered on top of the base move rules, using a flag-carrying **Move abstraction** to cleanly separate move generation from execution.
- Used a **simulate-move-then-check-for-check** approach to filter out moves that would leave the king exposed, and built **full undo support** by snapshotting state before each move.
- Validated correctness via **perft testing** against known reference values (**20 -> 4.86M nodes** across depths 1-5), sustaining **~170K nodes/sec** on single-threaded traversal.

Cache Simulator | *C++, STL, Hash Maps, Linked Lists*

- Built a configurable cache simulator supporting **LRU**, **LFU**, and **FIFO** eviction policies, with **O(1) get/put operations** using **hash maps** combined with **doubly linked lists** (LRU) and **per-frequency bucket** lists (LFU).
- Benchmarked all three policies against a locality-weighted access pattern (10,000 operations, 80/20 hot-cold key distribution), achieving up to **81.2% hit ratio with LFU** -- an **~7.5 point improvement over FIFO** (73.7%).
- Instrumented a stats-tracking layer reporting **hit ratio**, **miss ratio**, and **eviction counts** per policy, enabling direct quantitative comparison of eviction strategy trade-offs.

Mini Linux Filesystem | *C++, OOP, Trees*

- Built an in-memory, Linux-like filesystem in C++ supporting **mkdir**, **ls**, **cd**, **touch**, cp, mv, and rm, modeled as an **N-ary tree** with **Directory** and **File** nodes **inheriting** from an abstract **FSNode base**.
- Applied **inheritance** and **polymorphism** to unify file/directory operations under a common interface, and **composition** (unique_ptr-owned children) for automatic recursive cleanup on deletion.
- Designed a **unified path-resolution engine** handling absolute/relative paths, ./.. traversal, and parent+name extraction for not-yet-existing targets, reused across all seven commands.
- Implemented **cp via recursive deep-copy** and **mv via ownership transfer** (zero-copy rename/relocate), preserving correct parent-pointer and subtree integrity throughout.

ACHIEVEMENTS

- Secured **2nd Place** at the **Deutsche Börse Group Cybersecurity Hackathon** -- Performed forensic analysis in a Dockerized environment, uncovered **3 hidden flags**, and exploited an authentication vulnerability to extract target

data.

- Finished among the **Top 5** teams at the **KLA AI Hackathon** -- Built a SwinIR transformer-based super-resolution solution for semiconductor imaging that achieved **89.75% accuracy**, the highest in the competition.
- Achieved **Global Rank 1675** among **13,000+ participants** in the **IICPC CodeFest 2026 Global Prelims**.
- Solved **500+ coding problems** across Codeforces, LeetCode, GeeksforGeeks and InterviewBit.
- Achieved **Pupil** on **Codeforces** with a peak rating of **1372**.
- **Won 1st Place** at the national-level **Stemrobo Code for Health Hackathon** for developing an interactive voice-based yoga instructor application using block coding.

POSITIONS OF RESPONSIBILITY

Finance & Consulting Club, IIT Hyderabad

- **Web Coordinator:** Contributed to the design and development of club's website by building responsive, interactive frontend components.
- **Quant Coordinator**: One of the initial recruits of the club's newly launched Quant vertical; organized an alumni session connecting students with professionals in quant finance.
- **Consulting Associate**: Analyzed market sizing, profitability and operational strategy case studies; proposed a secure campus food-delivery solution.

EXTRACURRICULAR

- **Won multiple medals and trophies** at regional-level **chess** tournaments; represented at the zonal level.
- Represented the department in chess at **Diesta 2025,** IIT Hyderabad's Inter Department Sports Fest.